Class	Reg Number
17S	

Name ______________________________

MERIDIAN JUNIOR COLLEGE
JC2 Preliminary Examination
Higher 2

Calculator Model / No.

Chemistry

9729/04

Paper 4

11 September 2018

2 hours 30 minutes

READ THESE INSTRUCTIONS FIRST

Write your name, register number and class in the space provided above.

Give details of the practical shift and laboratory where appropriate, in the boxes provided.

Write in dark blue or black pen.

You may use an HB pencil for any diagrams or graphs.
Do not use correction fluid.

Answer **all** questions in the space provided on the Question Paper.

The use of approved scientific calculator is expected, where appropriate.

You may lose marks if you do not show your working or if you do not use appropriate units.

Qualitative Analysis Notes are provided on Pages **23** and **24**.
At the end of the examination, fasten all your work accurately together.

The number of marks is given in brackets [ ] at the end of each question or part question.

Shift
Laboratory

Examiner's Use	
Q1	/ 21
Q2	/ 19
Q3	/ 15
Total	/ 55

This document consists of **24** printed pages (including this cover page).

1 In this experiment, you are to determine the number of water of crystallisation (value of x) in an iron(II) salt, $(NH_4)_2Fe(SO_4)_2.\ x\ H_2O$ by titration with potassium manganate(VII).

You are provided with the following reagents.

FA 1 is potassium manganate(VII), $KMnO_4$ solution, of an approximate concentration 0.02 mol dm^{-3}
FA 2 is 0.0780 mol dm^{-3} ethanedioate ions $C_2O_4^{2-}$
FA 3 is a solution of the iron(II) salt containing 38.4 g dm^{-3} of $(NH_4)_2Fe(SO_4)_2.\ x\ H_2O$.
FA 4 is dilute sulfuric acid

Before using the provided solution of manganate(VII) ions, it is important to accurately determine its concentration. The exact concentration of **FA 1** can be determined with the given **FA 2** solution.

In an acidic medium, ethanedioate ions $C_2O_4^{2-}$ reacts with manganate(VII) ions MnO_4^- to form Mn^{2+} and CO_2 gas.

$$2MnO_4^- + 16H^+ + 5C_2O_4^{2-} \longrightarrow 2Mn^{2+} + 8H_2O + 10CO_2$$

As this reaction is very slow initially, MnO_4^- should only be added to a pre–heated $C_2O_4^{2-}$ solution in the presence of excess dilute acid. Subsequently, as the reaction proceeds, the rate of reaction will speed up and hence eliminating the need to continue heating.

Method

(a) (i) Standardisation of FA 1 solution

1. Fill the burette with **FA 1**.
2. Using a pipette, transfer 10.0 cm^3 of **FA 2** into the conical flask.
3. Using an appropriate measuring cylinder, transfer 20 cm^3 of **FA 4** to the same conical flask.
4. Heat this solution to about 65 °C using the Bunsen burner.
5. **Turn off your Bunsen burner.**

Caution: Use the heat–resistant foam to handle the neck of the heated flask.

6. Run **FA 1** from the burette into this flask until a **permanent** pale pink colour is obtained.

Initially, the colour of the **FA 1** will take some time to disappear. Subsequently, as the reaction proceeds, the rate of reaction will speed up

7. Record your titration results in the space provided on page **3**. Make certain that your recorded results show the precision of your working.
8. Repeat steps **1** to **7** as necessary until consistent results are obtained.

Results

(ii) From your titrations, obtain a suitable volume of **FA 1** to be used in your calculations. Show clearly how you obtained this volume.

[2]

volume of **FA 1** = ……………………………….

(iii) Determine the concentration of MnO_4^- in **FA 1**.

[1]

Concentration of MnO_4^- in **FA 1** = ……………………………….

M1	M2	M3

(iv) For the titration in **(a)(i)** between ethanedioate ions $C_2O_4^{2-}$ and manganate(VII) ions MnO_4^-, the ethanedioate solution needs to be at about 60 °C at the start.

However, during the titration, as **FA 1** is added, the temperature of the mixture decreases. This decrease in temperature did not cause the rate of reaction between $C_2O_4^{2-}$ and MnO_4^- added from the burette to decrease.

Suggest an explanation for this.

[1]

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....

....

M4	

(v) Calculate the maximum total percentage uncertainty for your titration in **(a)(i)** if the associated uncertainty associated with each reading using a 10.0 cm^3 pipette and a burette are ±0.03 cm^3 and ±0.05 cm^3 respectively.

Percentage uncertainty = ……………….....……. **[1]**

M5	

(b) **(i)** **Titration**

1. Fill the burette with **FA 1**.
2. Pipette 25.0 cm^3 of **FA3** into a conical flask and add 20 cm^3 of **FA 4** using a measuring cylinder.
3. Titrate **FA 3** with **FA 1** until the end–point is reached.
4. Carry out as many accurate titrations as you think necessary to obtain consistent results. Record your titration results in the space provided below.

(ii) From your titrations, obtain a suitable volume of **FA 1** to be used in your calculations. Show clearly how you obtained this volume.

[3]

volume of **FA 1** = ………………………………

M6	M7	M8

(c) Calculations

(i) The two half–equations for the reaction between the iron(II) salt and manganate(VII) ions are shown below.

$MnO_4^- + 8\ H^+ + 5\ e \longrightarrow Mn^{2+} + 4\ H_2O$

$Fe^{2+} \longrightarrow Fe^{3+} + e$

Calculate the amount of iron(II) ion present in 1.00 dm^3 of **FA 3**.

[1]

Amount of iron(II) ions in 1.00 dm^3 of **FA 3** = ……………………………….

M9	

(ii) Using your answer in **(c)(i)**, determine the value of x, the number of water of crystallisation in the iron(II) salt, $(NH_4)_2Fe(SO_4)_2.\ x\ H_2O$.

[A_r: Fe, 55.8; H, 1.0; N, 14.0; O, 16.0; S, 32.1]

[3]

x = ……………………………….

M10	M11	M12

(d) In the titration of the iron(II) salt and manganate(VII) ion in **(b)(i)**, it was observed that the colour of the mixture in the conical flask gradually changes before an excess drop of manganate(VII) turns the mixture pale pink at the end point.

State and explain the colour changes you observed in the titration, in terms of the complex ions involved.

[2]

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....

M13	M14

(e) Planning

A saturated aqueous solution of magnesium methanoate, $Mg(HCOO)_2$, has a solubility of approximately **150 g dm^{-3}** at room temperature. Its exact solubility can be determined by titrating magnesium methanoate against aqueous potassium manganate(VII).

During the titration, methanoate ion, $HCOO^-$ reacts with manganate(VII) ion, MnO_4^- in an acidic medium as shown in the equation below.

$$2MnO_4^- + 5HCOO^- + 11H^+ \longrightarrow 2Mn^{2+} + 5CO_2 + 8H_2O$$

You may assume that you are provided with:

- 100 g of solid magnesium methanoate, $Mg(HCOO)_2$
- 0.0200 mol dm^{-3} aqueous potassium manganate(VII), $KMnO_4$
- 1.00 mol dm^{-3} sulfuric acid, H_2SO_4
- the apparatus and equipment normally found in a school or college laboratory.

In order to obtain a reliable and feasible titre value, the saturated solution of magnesium methanoate needs to be diluted accurately (**~50 times** dilution) before titration.

(i) Plan an investigation to determine the effect of temperature on the solubility of magnesium methanoate, $Mg(HCOO)_2$. Your plan must enable you to plot a graph that includes data from the experiment.

Your plan should include details of:

- the preparation of 50 cm^3 of **saturated** aqueous solutions of $Mg(HCOO)_2$ at different temperatures
- the preparation of diluted solutions of $Mg(HCOO)_2$ for titration
- an outline of how the titration could be carried out
- a sketch of the graph you would expect to obtain
- an outline how the results can be used to determine the effect of temperature on the solubility of $Mg(HCOO)_2$

[6]

YOU ARE NOT REQUIRED TO PERFORM THIS EXPERIMENT.

M15 to M20

(ii) A student used the same titration method, this time to measure the concentration of a saturated solution of barium methanoate. Explain why the acidification of the solution with dilute sulfuric acid might make the titration difficult to do.

[1]

M21	

[Total: 21]

2 Strong acids, such as hydrochloric acid, HC*l*, are completely ionised in aqueous solution. Weak acids, such as ethanoic acid, CH_3COOH, are partially ionised in aqueous solution.

You will investigate the enthalpy change for the reaction of an excess of each of these acids with magnesium and hence determine the energy needed to cause the weak acid to ionise completely.

(a) **Experiment 1**

FA 5 is 1.0 mol dm^{-3} hydrochloric acid, HC*l*.
FA 6 is magnesium, Mg.

1. Weigh the magnesium strip, **FA 6**, and record the mass reading.
2. Support the Sytrofoam cup in the 250 cm^3 beaker.
3. Coil the magnesium strip loosely so that it fits into the bottom of the Sytrofoam cup and then remove the strip.
4. Use the measuring cylinder to transfer 25 cm^3 of the acid, **FA 5**, into the Sytrofoam cup.
5. Place the thermometer in the acid and read the initial temperature. This is the temperature at time zero (t = 0).
6. Start timing and do not stop the stop watch until the whole experiment has been completed.
7. Read the temperature of the acid every half minute for two minutes.
8. At time t = 2.5 minutes drop the magnesium strip, **FA 6**, into the acid and stir the mixture. **Do not try to read the temperature at this time**.
9. Measure and record, the temperature of the mixture at t = 3.0 minutes and then every half minute until t = 8.0 minutes. Stir the mixture continuously between thermometer readings.
10. Rinse the Sytrofoam cup for use in **Experiment 2**. Shake to remove excess water.

In the space provided below, record all measurements of mass. Record all your results of temperature and time in a table.

Results

[4]

M1	M2	M3	M4

(b) Plot a graph of temperature on the y–axis against time on the x–axis on the grid below. The scale for the temperature axis should extend 10 °C greater than the maximum temperature you recorded. You will use the graph to determine the theoretical maximum temperature rise at 2.5 minutes.

Draw two lines of best fit through the points on your graph, the first for the temperature before adding **FA 6** and the second for the cooling of the mixture once the reaction is complete.

Extrapolate the two lines to 2.5 minutes, draw a vertical line between the two and determine the theoretical rise in temperature at this time.

theoretical rise in temperature at 2.5 minutes = ... **[4]**

M5	M6	M7	M8

(c) Magnesium reacts with hydrochloric acid according to the equation shown.

$$Mg\ (s) + 2HCl\ (aq) \longrightarrow MgCl_2\ (aq) + H_2\ (g)$$

(i) Use your answer to **(b)** to calculate the heat energy, in joules, given out when **FA 6** is added to the acid.
[Assume 4.2 J of heat energy raises the temperature of 1.0 cm^3 of the mixture by 1.0 °C.]
[1]

heat energy evolved = ...

M9	

(ii) Calculate the enthalpy change when 1 mole of **FA 6**, Mg, reacts with hydrochloric acid. [A_r: Mg, 24.3]
[1]

enthalpy change, ΔH = ...

M10	

(d) **(i)** Calculate the percentage error in the measurement of ΔT in the experiment.
[1]

M11	

(ii) A student suggested that the experiment carried out in **(a)** could be improved by using a catalyst.

Would the use of a catalyst improve the accuracy of the results in this experiment? Give a reason for your answer.

[1]

M12	

(iii) Another student could not find the hydrochloric acid, **FA 5**, and used sulfuric acid, H_2SO_4, instead. He used the same volume and the same concentration as the hydrochloric acid in **FA 5**.

What effect would this change have on the temperature rise in the experiment? Give a reason for your answer.

[1]

M13	

(e) **Experiment 2**

FA 7 is 1.0 mol dm^{-3} ethanoic acid, CH_3COOH
FA 8 contains one strip of magnesium, Mg of a mass of 0.087 g

Read the whole **Experiment 2** before starting any practical work.

1. Support the Sytrofoam cup in the 250 cm^3 beaker.
2. Coil the magnesium strip loosely so that it fits into the bottom of the Sytrofoam cup and then remove the strip.
3. Use the measuring cylinder to transfer 25 cm^3 of the acid, **FA 7**, into the Sytrofoam cup.
4. Place the thermometer in the acid and measure and record the initial temperature of the acid.
5. Add the piece of magnesium, **FA 8**, into the acid in the cup.
6. Using the thermometer, stir the mixture continuously until it reaches its maximum temperature. This could take several minutes. Record this temperature, T_{max}.
7. Calculate and record the change in temperature, ΔT.
8. Complete **Table 2.1** by recording all measurements of temperature.

Table 2.1

Initial temperature / ^{o}C	
Maximum temperature, T_{max} / ^{o}C	
Change in temperature, ΔT / ^{o}C	

[1]

M14	

Magnesium reacts with ethanoic acid according to the equation shown.

$$Mg\ (s) + 2CH_3COOH\ (aq) \longrightarrow Mg(CH_3COO)_2\ (aq) + H_2\ (g) \quad \Delta H_1$$

Use your results in **Table 2.1** to calculate the enthalpy change, ΔH_1, when 1 mole of Mg, **FA 8**, reacts with excess ethanoic acid. Assume 4.2 J of heat energy changes the temperature of 1.0 cm^3 of the mixture by 1.0 °C. [A_r: Mg, 24.3]

[1]

ΔH_1 = ...

M15	

(f) Hydrochloric acid is a strong acid; it is fully ionised in aqueous solution. When the same experiment in **(e)** was repeated with hydrochloric acid, **FA 5**, instead of ethanoic acid, the enthalpy change, ΔH_2, when 1 mole of Mg reacts with excess hydrochloric acid was determined to be – 461.1 kJ mol^{-1}.

Ethanoic acid is a weak acid. It is partially ionised in aqueous solution.

$$CH_3COOH\ (aq) \rightleftharpoons CH_3COO^-\ (aq) + H^+\ (aq)$$

You are to determine the energy needed for one mole of ethanoic acid to ionise completely.

$$CH_3COOH\ (aq) \longrightarrow CH_3COO^-\ (aq) + H^+\ (aq) \qquad \Delta H_3$$

Use the values for the enthalpy changes, ΔH_1 you obtained in **(e)** and the ΔH_2 provided to calculate the energy change for the ionisation of ethanoic acid, ΔH_3.

Show clearly how you obtained your answer.

[2]

ΔH_3 = ...

M16	M17

(g) The experiment in **(e)** was repeated using trichloroethanoic acid instead of ethanoic acid.

$Mg(s) + 2\,CCl_3COOH\,(aq) \longrightarrow Mg(CCl_3COO)_2\,(aq) + H_2\,(g)$ ***reaction 3***

Trichloroethanoic acid, CCl_3COOH, is a weak acid that is however stronger than ethanoic acid. The enthalpy change for ***reaction 3*** is between the two values given in **Table 2.2**.

Table 2.2

reaction	equation	ΔH / kJ mol^{-1}
1	$Mg\,(s) + 2CH_3COOH\,(aq) \longrightarrow Mg(CH_3COO)_2\,(aq) + H_2\,(g)$	−460.3
2	$Mg\,(s) + 2HCl\,(aq) \longrightarrow MgCl_2\,(aq) + H_2\,(g)$	−464.1

(i) Explain why the enthalpy change for reaction 3 is more exothermic than the enthalpy change for reaction 1.

[1]

....

....

....

....

....

M18	

(ii) Explain why the enthalpy change for reaction 3 is less exothermic than the enthalpy change for reaction 2.

[1]

....

....

....

....

M19	

[Total: 19]

3 Qualitative Analysis

At each stage of any test you are to record details of the following.

- colour changes seen
- the formation of any precipitate
- the solubility of such precipitates in an excess of the reagent added

Where reagents are selected for use in a test, the **name** or **correct formula** of the element or compound must be given.

Where gases are released they should be identified by a test, **described in the appropriate place in your observations**.

You should indicate clearly at what stage in a test a change occurs.
No additional tests for ions present should be attempted.

(a) **(i)** **FA 9**, **FA 10** and **FA 11** each contain one anion and one cation.

Carry out the following tests and record your observations.

Test	***observations***		
	FA 9	**FA 10**	**FA 11**
To a 1 cm depth of solution in a test–tube, add a few drops of aqueous silver nitrate, then			
add aqueous ammonia.			
To a 1 cm depth of solution in a test–tube, add a few drops of aqueous barium nitrate, then			
add dilute nitric acid.			

Test	*observations*		
	FA 9	**FA 10**	**FA 11**
To a 1 cm depth of solution in a test–tube, add a spatula of solid sodium carbonate.			

M1	M2	M3	M4

(ii) What cation is present in **FA 9**, **FA 10** and **FA 11**?

....

(iii) Suggest another chemical test that you could carry out to confirm the presence of the cation you identified in **(ii)**.

Carry out this test on one of **FA 9**, **FA 10** and **FA 11** and record your observations.

Test: ....

....

....

Observations: ....

....

....

(iv) Complete the table to identify, as far as possible, the anions present in **FA 9**, **FA 10** and **FA 11**. If you are not able to identify the anion from the tests you carried out in **(i),** write 'unknown'.

	FA 9	**FA 10**	**FA 11**
ion present			

M5	M6	M7	M8

(v) For any one anion that you were unable to identify in **(iv)** you are to devise a test or tests that will enable you to identify it. You can assume that it is one of the anions listed in the Qualitative Analysis Notes.

Carry out the test(s), record the observation(s) you obtained and identify the unknown anion.

In the space below, draw a single table to record details of the test(s) performed and observations made.

Identify the anion and state the evidences for your identification.

Anion: ....

Evidence: ....

....

....

....

[10]

M9	M10

(b) **FA 12** is an aqueous solution of a mixture containing two anions.
Carry out the following tests and record your observations.

Test	*Observations*
To a 1 cm depth of **FA 12** in a test–tube, add a 1 cm depth of dilute hydrochloric acid, then	
add a few drops of hydrogen peroxide, then	
add a few drops of starch.	
To a 1 cm depth of **FA 12** in a test–tube, add a 3 cm depth of aqueous copper(II) sulfate, then	
add a 1 cm depth of dilute hydrochloric acid, then	
add aqueous sodium thiosulfate.	

From these observations, identify the two anions present in **FA 12**.

Anions present in **FA 12** .... and ....

[5]

M11	M12	M13	M14	M15

[Total: 15]

9 Qualitative Analysis Notes
[ppt. = precipitate]

9(a) Reactions of aqueous cations

cation	*reaction with*	
	$NaOH(aq)$	$NH_3(aq)$
aluminium, $Al^{3+}(aq)$	white ppt. soluble in excess	white ppt. insoluble in excess
ammonium, NH_4^+ (aq)	ammonia produced on heating	–
barium, $Ba^{2+}(aq)$	no ppt. (if reagents are pure)	no ppt.
calcium, $Ca^{2+}(aq)$	white. ppt. with high $[Ca^{2+}(aq)]$	no ppt.
chromium(III), $Cr^{3+}(aq)$	grey-green ppt. soluble in excess giving dark green solution	grey-green ppt. insoluble in excess
copper(II), $Cu^{2+}(aq)$,	pale blue ppt. insoluble in excess	blue ppt. soluble in excess giving dark blue solution
iron(II), $Fe^{2+}(aq)$	green ppt., turning brown on contact with air insoluble in excess	green ppt., turning brown on contact with air insoluble in excess
iron(III), $Fe^{3+}(aq)$	red-brown ppt. insoluble in excess	red-brown ppt. insoluble in excess
magnesium, $Mg^{2+}(aq)$	white ppt. insoluble in excess	white ppt. insoluble in excess
manganese(II), $Mn^{2+}(aq)$	off-white ppt., rapidly turning brown on contact with air insoluble in excess	off-white ppt., rapidly turning brown on contact with air insoluble in excess
zinc, $Zn^{2+}(aq)$	white ppt. soluble in excess	white ppt. soluble in excess

9(b) Reactions of anions

anion	*reaction*
carbonate, CO_3^{2-}	CO_2 liberated by dilute acids
chloride, Cl^-(aq)	gives white ppt. with Ag^+(aq) (soluble in NH_3(aq))
bromide, Br^-(aq)	gives pale cream ppt. with Ag^+(aq) (partially soluble in NH_3(aq))
iodide, I^-(aq)	gives yellow ppt. with Ag^+(aq) (insoluble in NH_3(aq))
nitrate, NO_3^- (aq)	NH_3 liberated on heating with OH^-(aq) and Al foil
nitrite, NO_2^- (aq)	NH_3 liberated on heating with OH^-(aq) and Al foil; NO liberated by dilute acids (colourless NO $\rightarrow$ (pale) brown NO_2 in air)
sulfate, SO_4^{2-} (aq)	gives white ppt. with Ba^{2+}(aq) (insoluble in excess dilute strong acids)
sulfite, SO_3^{2-} (aq)	SO_2 liberated with dilute acids; gives white ppt. with Ba^{2+}(aq) (soluble in dilute strong acids)

9(c) Tests for gases

gas	*test and test result*
ammonia, NH_3	turns damp red litmus paper blue
carbon dioxide, CO_2	gives a white ppt. with limewater (ppt. dissolves with excess CO_2)
chlorine, Cl_2	bleaches damp litmus paper
hydrogen, H_2	"pops" with a lighted splint
oxygen, O_2	relights a glowing splint
sulfur dioxide, SO_2	turns aqueous acidified potassium manganate(VII) from purple to colourless

9(d) Colour of halogens

halogen	*colour of element*	*colour in aqueous solution*	*colour in hexane*
chlorine, Cl_2	greenish yellow gas	pale yellow	pale yellow
bromine, Br_2	reddish brown gas / liquid	orange	orange-red
iodine, I_2	black solid / purple gas	brown	purple